Anishka Chawla

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Professional Profile

Computer Science and Engineering student with a strong foundation in software architecture, data structures, and algorithms. Proven track record of designing, testing, and deploying high-quality applications and driving end-to-end development.

Education

Thapar Institute Of Engineering and Technology — Patiala, Punjab
B.E. in Computer Engineering, CGPA-8.12 — *Expected - 2027*

West Point School — Kotakapura, Punjab
CBSE-XII, Percentage-89% — *2023*

DAV Public School — Ludhiana,Punjab
CBSE-X, Percentage-94% — *2021*

Experience

Research Intern — June 2026 – July 2026
CSIR - Central Scientific Instruments Organisation (CSIO) — *Chandigarh*
- Designed and built deep learning-based regression models for hyperspectral millet nutrient prediction, leveraging analytics to evaluate reduced wavelength feature sets.
- Benchmarked a custom hybrid architecture against existing models through rigorous testing to ensure reliability and accurate multi-output estimation of moisture, fats, carbohydrates, and protein.

Intern — June 2025 – July 2025
Colt Technology Services — *Remote*
- Developed a **Semantic Vector Search Engine** utilizing **Streamlit**, **Google Cloud BigQuery**, and **Vertex AI** enabling search across datasets with 10K+ records.
- Engineered a robust **highlighting mechanism** leveraging **BigQuery ML embeddings** that improved search result relevance by 30% and enhanced user experience.
- Contributed to data-driven decision making by integrating **PowerBI** dashboards for analytics and reporting

Projects

Semantic ATS Resume Matching System | ***Python, FastAPI, React.js, Tailwind CSS, Sentence Transformers, NLP*** | Link
- Built a Semantic ATS system to evaluate resume–job description fit using keyword matching, semantic similarity, and structural analysis, generating an explainable 0–100 ATS score with matched and missing skill insights.
- Developed the end-to-end application with FastAPI, React, sentence-transformers, and Gemini API, enabling resume/JD parsing, skill-gap analysis, and AI-generated feedback for resume optimization.

Autonomous Multi-Agent Developer Suite | ***CrewAI, Python, MCP, Docker, Gemini API, LLMs*** | Link
- Built an autonomous multi-agent system using CrewAI and LLMs, engineering role-based AI hierarchies (PM, Developer, QA) to automate requirement analysis and zero-intervention code synthesis.
- Integrated a containerized Docker sandbox with structured task-handoff protocols to securely compile, test, and iteratively debug AI-generated code within an isolated execution environment.

LinkedIn Personal Brand Builder | ***React.js, Node.js, MongoDB, Python, Gemini API, Firebase Auth*** | Link
- Developed an AI-powered web application that automatically scrapes 100+ industry news articles, performs intelligent article classification, and generates concise AI-based summaries using integrated generative AI models to streamline professional content discovery.
- Integrated advanced generative AI APIs to convert summarized news into polished, LinkedIn-ready post drafts, enabling automated personal branding in <5 seconds per post.

Technical Skills

Languages: Python, C, C++, HTML, CSS, JavaScript
Generative AI and Agentic AI: LangChain, RAG, CrewAI, OpenAI Agents SDK
Development Frameworks: React.js, Node.js, Next.js, Tailwind CSS
Cloud Platforms: Google Cloud Platform(GCP)
Tools and Software: Power BI, MongoDB, Visual Studio Code, GitHub
Core CS Concepts: Data Structures and Algorithms, OOPs, Operating Systems, SQL, Computer Networks

Certifications

- **AI Engineer Agentic Track: The Complete Agent and MCP Course** | Link